

Single molecule processes on the stepwise movement of ATP-driven molecular motors

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Abstract

Movement is a fundamental characteristic of all living things. This biogenic function that is attributed to the molecular motors such as kinesin, dynein and myosin. Molecular motors generate forces by using chemical energy derived from the hydrolysis reaction of ATP molecules. Despite a large number of studies on this topic, the chemomechanical energy transduction mechanism is still unsolved. In this study, we have investigated the chemomechanical coupling of the ATPase cycle to the mechanical events of the molecular motor kinesin using single molecule detection (SMD) techniques. The SMD techniques allowed to detection of the movement of single kinesin molecules along a microtubule and showed that kinesin steps mainly in the forward direction, but occasionally in the backward. The stepping direction is determined by a certain load-dependent process, on which the stochastic behavior is well characterized by Feynman's thermal ratchet model. The driving force of the stepwise movement is essentially Brownian motion, but it is biased in the forward direction by using the free energy released from the hydrolysis of ATP.

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1. Introduction

Molecular motors are biogenic force generators acting in the nanometer range. Linear motors pro-

duce sliding movements along filamentous structures called protein tracks; for example, myosin slides along an actin filament (Finer et al., 1994; Ishijima et al., 1994), kinesin (Block et al., 1990; Vale et al., 1996) and dynein (Sakakibara et al., 1999) along a microtubule. Many of the molecular motors have the ability to rotate; for example, the bacterial flagellar motor (Silverman and Simon, 1974; Kudo et al., 1990), and F₀F₁-ATP synthase (Noji et al., 1997).

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In addition, RNA polymerase causes the rotation of DNA utilizing a sliding motion during transcription (Harada et al., 2001). These movements and forces generated by the motors play essential roles in cellular functions.

The linear motor kinesin is widely distributed in almost all eukaryotic cells and the sliding motion along a microtubule results in intracellular vesicle transport and cell division (Hirokawa, 1998). This motor has two motor domains called heads, each of which includes the ATP- and microtubule-binding sites (Kull et al., 1996). Kinesin moves toward the plus end of a microtubule over long distances without dissociation. The movement of single kinesin molecules can be directly observed using single molecule detection (SMD) techniques. Kinesin molecules attached to the single probes such as microspheres (Block et al., 1990), and fluorophores (Vale et al., 1996; Yajima et al., 2002), can be observed sliding along microtubules. Optical trapping nanometry is a highly advanced SMD technique that can detect the displacements of single molecules with a resolution of nanometers. Results using this technique showed that single kinesin molecules produce stepwise displacements with a size of 8 nm (Svoboda et al., 1993).

Kinesin steps mainly in the forward direction (to the plus end of the microtubule), but occasionally in the backward direction (to its minus end) (Svoboda and Block, 1994). In previous studies, the stepping processes were investigated using the averaged time courses of the movement. Both the forward and backward steps were included in the traces, but individual backward steps were ignored in the analysis being considered rare events when compared with the regular 8-nm steps in the forward direction. The mechanism underlying these movements in the forward and backward directions remains unclear.

In this study, we investigated the stepping mechanism by correlating the hydrolysis of ATP to the forward and backward movements of single kinesin molecules. If the hydrolysis of single ATP molecules always results in 'successive' 8-nm steps in the forward direction, then the hydrolysis of ATP can be said to be tightly coupled to the mechanical event and the chemomechanical energy transduction process would be deterministic. However, our results do not sup-

port this idea. Alternatively, the hydrolysis of a single ATP molecule may be coupled to either the forward or backward movement, indicating a loosely coupled model (Oosawa, 2000). This stochastic behavior of kinesin motors can be well explained by a biased Brownian model, such as the one-dimensional Brownian motion on the asymmetric potential, and the thermal ratchet mechanism (Feynman, 1963; Astumian, 1997; Reimann, 2002). The movement of molecular motors is most likely due to Brownian motion that is biased in the forward direction utilizing chemical energy of ATP molecules.

2. Single molecule enzymology of the molecular motor, kinesin

2.1. Development of dark-field style optical trapping nanometry

The previous optical trapping nanometry techniques could not resolve the individual stepwise movements under several experimental conditions such as near zero loads and at saturating concentrations of ATP, where successive ATP hydrolysis occurs very rapidly. To detect such stepwise movements, we improved the temporal and positional resolution of the optical trapping nanometry. The temporal resolution of the experimental system is primarily limited by the viscous drag of the bead. To overcome this problem, we attached kinesin to beads, 0.2 μm in diameter, instead of 1 μm . Small beads are advantageous because they show a better response time. However, the images produced from small beads on a detector are not clear, and the signal-to-noise ratio for nanometry is low. To overcome this problem, we used dark-field illumination (Nishiyama et al., 2001) and the optical system was built into a conventional optical trapping nanometry system (Kojima et al., 1997). The bead was illuminated diagonally by a focused red laser and its dark-field image was projected onto a quadrant-photodiode (Fig. 1a). By measuring the differential outputs, we could detect the nanometer displacements of the molecular motor on the bead. The laser dark-field illumination technique was also used to detect the high speed motion of molecular motors with high signal-to-noise ratio as demonstrated in flagella rotation for bacteria (Kudo et al., 1990) and

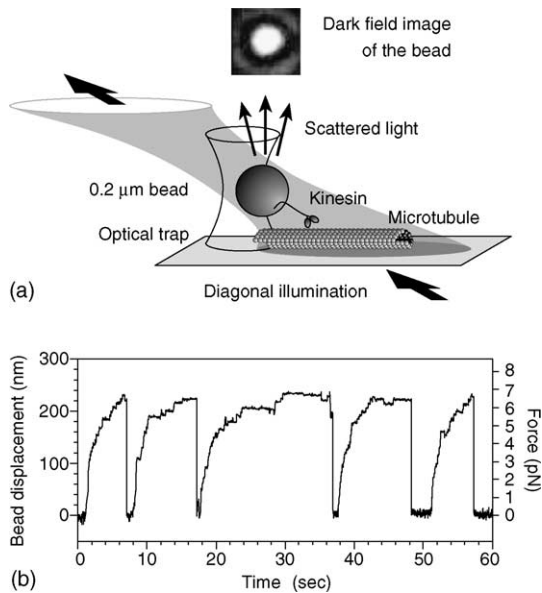


Fig. 1. An *in vitro* motility assay of single kinesin molecules. (a) Schematic diagram of the optical trapping nanometry system (not to scale). Nanometer – displacements and piconewton – forces of kinesin were measured. (b) Displacement of a kinesin-coated bead interacting with a microtubule at $10\ \mu\text{M}$ ATP. The force produced by kinesin was calculated from the displacement of a bead multiplied by the trap stiffness ($=0.03\ \text{pN/nm}$). The bead displacement was caused by single kinesin molecules. Confirmation that these forces were produced by single molecules came from the ratio of the moving beads coated by kinesin and maximum force produced (Kojima et al., 1997). Assay conditions: $80\ \text{mM}$ PIPES, $1\ \text{mM}$ EGTA, $2\ \text{mM}$ MgCl_2 , $1\ \text{mM}$ and $10\ \mu\text{M}$ ATP, $25 \pm 1^\circ\text{C}$.

rotational substeps in the 120° step for $\text{F}_1\text{-ATPase}$ (Yasuda et al., 2001).

2.2. Stepwise movements of single kinesin molecules

A bead attached to a single kinesin molecule in solution was trapped by an infrared laser beam and brought into contact with a microtubule fixed on a glass surface. When kinesin associated with a microtubule, the bead was rapidly displaced from the trapping center and then the velocity decreased until it reached a plateau (Fig. 1b). As the bead moved from the trap center, the trap force linearly increased with the distance from the trap center. Kinesin moved against the load as a result of the trapping force. The bead stayed at an almost constant plateau level for a certain period of time and then returned to the trap center abruptly.

This force level was considered to be the maximum force generated by the single kinesin molecules. The force of kinesin was calculated from the bead displacement and the trapped stiffness with sub-piconewton accuracy (Kojima et al., 1997).

The time course of the kinesin movement showed a distinctive stepwise pattern with regular 8-nm steps, which was clearly demonstrated by the lines drawn at 8 nm intervals in Fig. 2a. Kinesin moves primarily in the forward direction but occasionally moves in the backward direction (see arrows in Fig. 2b). The backward step size was either 8 or 16 nm, different from the steps in the forward direction. The 16-nm step could not be resolved into two separate 8-nm steps, suggesting that the back steps may result from the re-binding of kinesin to other sites of microtubule after detachment. A step size of 8 nm agrees completely with the length of an $\alpha\beta$ -tubulin dimer on the microtubule, strongly suggesting that the step size is likely to be determined by the structural periodicity of microtubule.

The 8-nm step results from the hydrolysis of single ATP molecules (Schnitzer and Block, 1997; Hua et al., 1997; Visscher et al., 1999; Nishiyama et al., 2002). It has been speculated that a single step is composed of many separate intricate motions. However, experimental evidence for such motions have not been observed. This problem is one of the “hot topics” for researchers studying the stepping mechanism (Schief and Howard, 2001). To answer this question, we have tested if other short steps exist within the dwell time between the 8-nm steps. No significant displacements were found both in the raw and averaged traces (Fig. 2d). Further, we analyzed the rising phase of the 8-nm steps with a microseconds time scale and results demonstrated that the 8-nm step is composed of sequential 4-nm substeps (Fig. 2e). The time course was of $50\ \mu\text{s}$, which is less than 1% of ATP turnover time (10 ms) at near zero loads. This implies that kinesin is almost ‘resting’ on the microtubule. In the following sections, this resting time has been characterized and related to the hydrolysis processes of ATP molecules.

2.3. Stepping kinetics of the forward and backward movements

In previous studies, the backward movements were included in the kinesin displacements but were ignored

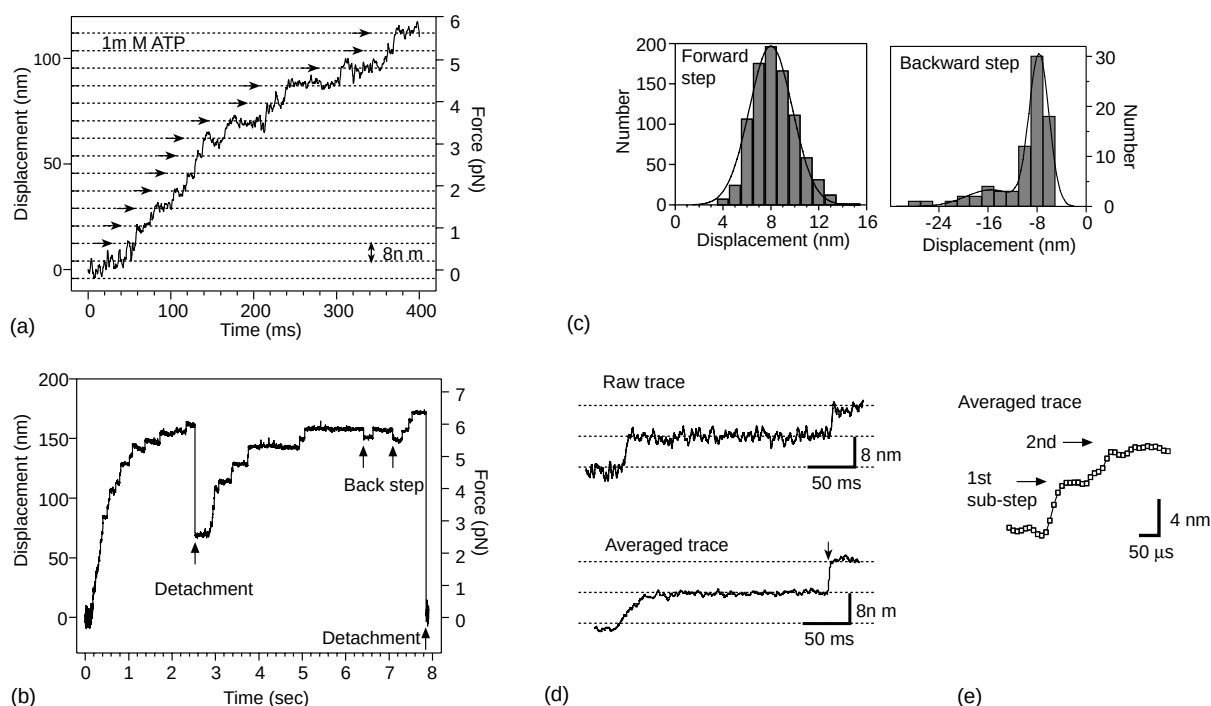


Fig. 2. Stepwise movements caused by single kinesin molecules. Time courses of the displacements of single kinesin molecules. [ATP] = 1 mM (a) and 10 μ M (b). The displacement of kinesin was obtained from the bead displacement taking into consideration the compliance of the experimental system. The force was calculated from the bead displacement and the trap stiffness ($=0.05$ pN/nm in (a)) (right hand ordinate). (c) The step size in the forward (left) and backward direction (right) is independent of the load and concentrations of ATP. (Left) 1 mM ATP. The histogram was fitted by a Gaussian curve with 8.0 ± 1.8 nm (mean $\pm$ S.D.). (Right) The displacement data (<30 nm) at ATP concentrations of 1 mM and 10 μ M have been included. The solid line is the sum of two Gaussian peaks fitted to the data; the positions and widths of each peak were -7.7 and 1.6 nm, -15.5 and 4.1 nm (twice the displacement at the main peak), respectively. (d) The raw and averaged displacements within the dwell time. [ATP] = 10 μ M. Upper trace is a typical trace, which was recorded with a bandwidth of 500 Hz. The lower trace was obtained by averaging the 15 traces (dwell time = 200 ± 20 ms) at the 4-nm point of the “end” step (arrow). (e) The sequential 4-nm substeps. Data were recorded at a sampling of 100 kHz after passing through fourth-order Butterworth low-pass filter at 20 kHz. Trace shows the averaged data of the rising phase of the 8-nm step, for which the rise times were 50–100 μ s (Nishiyama et al., 2001).

in the analysis because the back steps were considered to be rare events compared with the regular 8-nm step in the forward direction. In this study, we explored the whole stepping mechanism resulting in the forward and backward movements. We analyzed the force, stepping directionality and dwell time at saturating and limiting ATP concentrations (Nishiyama et al., 2002).

Increasing in load either the occurrence of the backward movements increased, or the ratio of the forward to the backward movements decreased (Fig. 3a), indicating that the stepping direction is determined at a certain load-dependent process. To

study this load-dependent process, we analyzed the load-dependence of the dwell time between the adjacent stepwise movements. Fig. 3b shows the average dwell time before the forward movement (8-nm step). The dwell time increased with load and decreased with a concentration of ATP. Interestingly, similar results were obtained for movements in the backward direction, that is, back step and detachment. Previous studies showed that the generation process of the 8-nm step for kinesin could be well explained by a 3-state model, no nucleotide [K], ATP [K.T] and ADP bound states [K.D] (Howard, 2001; Thomas et al., 2001). Here, the kinetic model has been expanded by

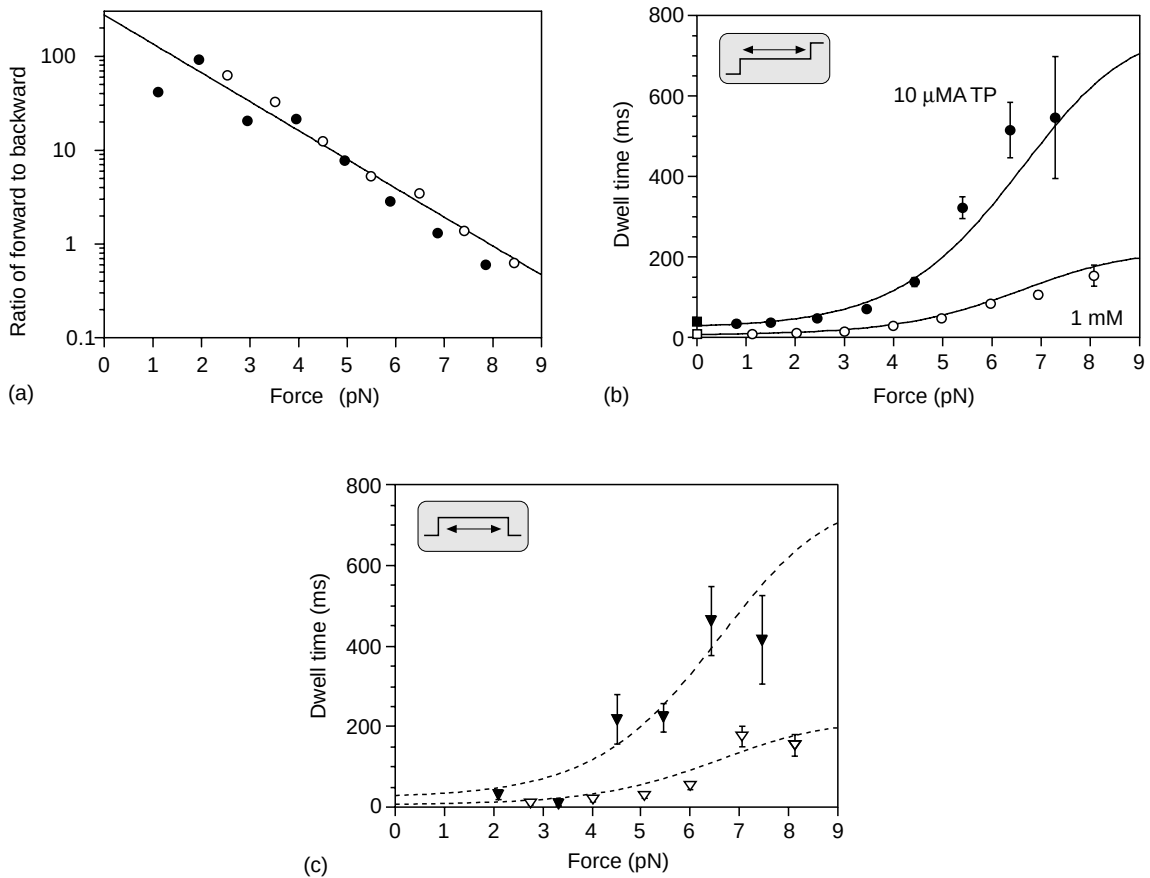


Fig. 3. Analysis of the forward and backward movements. (a) Ratio of the steps in the forward to backward directions. The plots at 1 mM ATP ($\circ$, $n = 71$ –261, total = 1464) and 10 μ M ATP ($\bullet$, $n = 8$ –186, 990) were fitted to Eq. (1) with $R_0 = 221$ and $d = 2.9$ nm. (b) Dwell time before the forward movement as a function of force. Each point represents the average dwell time at ATP concentrations of 1 mM ($\circ$, $n = 34$ –222, 1183) and 10 μ M ($\bullet$, $n = 13$ –177, 889). The plots have been fitted by Eqs. (3) and (4) with $\tau_2 = 7.0$ ms, $\tau_3 = 1.3$ ms and $d_f = 2.9$ nm. (c) Dwell time before the backward movement. Each plot is the average value of the dwell time at ATP concentrations of 1 mM (∇ , $n = 6$ –48, 168) and 10 μ M ($\blacktriangledown$, $n = 7$ –37, 107). The dotted curves were the same as the solid lines in (b), and are shown here for comparison.

including the backward movements as an additional pathway (Fig. 4a).

The forward and backward movements occurred in parallel in the [K.D] state, with the rate constants of k_{3f} and k_{3b} , respectively. The sum of the movements in the forward and backward directions gives a rate constant from the states [K.D] to [K]. The ratio of the number of movements in the forward to backward directions is equal to k_{3f}/k_{3b} . According to Arrhenius/Eyring kinetics, the rate constant is related to the activation energy of the barrier potential. The asymmetric potential was,

therefore, used to analyze the bidirectional stepwise movements.

The activation energy of the forward and backward directions can be described by $E_f + Fd_f$ and $E_b + Fd_b$, respectively, where E_f and E_b are the heights of the barrier maximum at zero load, and d_f and d_b are the characteristic distances against the load of F (Fig. 4b). If the Boltzmann energy distribution is assumed, then the rates in the forward and backward direction should be proportional to $\exp(-(E_f + Fd_f)/k_B T)$ and $\exp(-(E_b + Fd_b)/k_B T)$, respectively, where k_B is the Boltzmann constant and T is the experimental

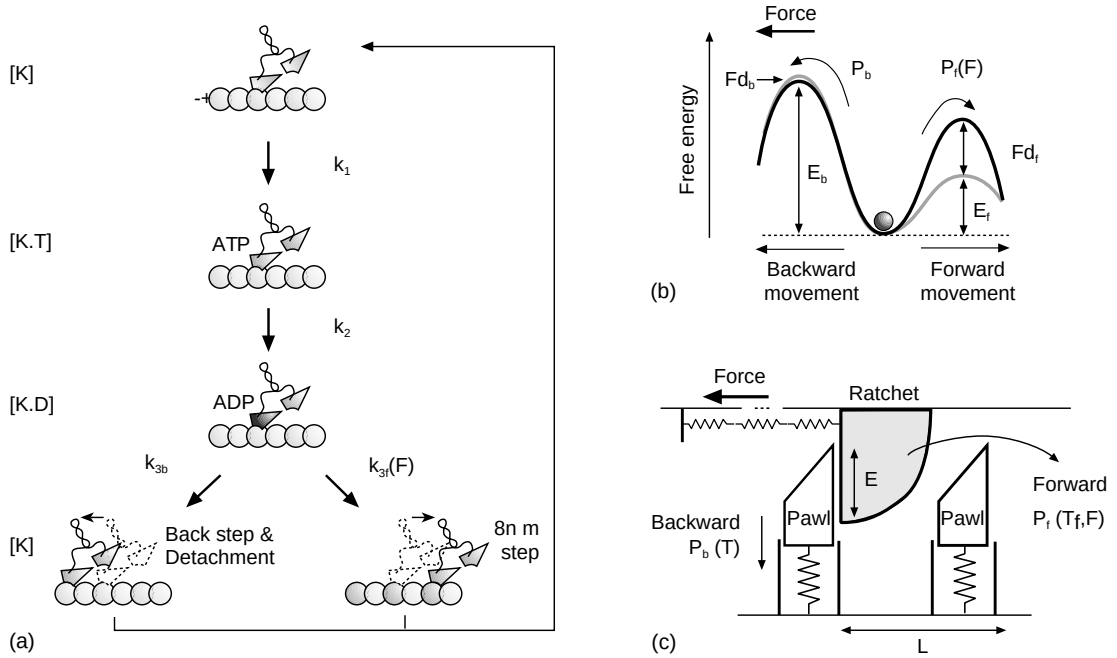


Fig. 4. Stepping kinetics of the bidirectional movements. (a) The branched kinetic pathway. Kinesin attached to the microtubule with one motor domain, [K]. The ATP molecule binds to the head, [K.T] and is hydrolyzed, [K.D]. After kinesin moved to either the forward or backward direction, the ADP molecules were released [K]. (b) Asymmetric potential of the free energy. The probabilities of the movements in the forward and backward directions have been determined by the different activation energies, respectively. (c) Feynman's thermal ratchet model. The ratchet and pawl are isolated at different temperatures of T_f and T_b ($=T$), respectively. The probabilities of the movements in the forward and backward directions are determined by a difference in temperatures. See text for details.

temperature (298 K). The number of movements in the forward direction relative to those in the backward direction is equal to k_{3f}/k_{3b} under kinetic control and can be described by,

$$\frac{k_{3f}}{k_{3b}} = R_0 \exp\left(-\frac{Fd}{k_B T}\right), \quad (1)$$

$$R_0 = \exp\left[\frac{E_b - E_f}{k_B T}\right], \quad (2)$$

where d is the difference of the characteristic distances ($=d_f - d_b$). The plots in Fig. 3a were fitted by Eq. (1) with $R_0 = 221$ and $d = 2.9$ nm. Thus, the energy difference between barrier maxima in the forward and backward directions was $5.4k_B T$ at zero load. As the load increased, the energy difference decreased linearly increasing the probability of backward movements. When the load reached 7.6 pN, the barrier height of the forward direction was equal

to that in the backward, resulting in the disappearance of the directional motion. Thus, the stall force could be described as the force where the probabilities of the forward and backward movements are equal.

According to our branched 3-state model, the ATP turnover time, τ_{cat} , can be described by the inverse of k_2 and the inverse of $k_{3f} + k_{3b}$. Thus, τ_{cat} is given by,

$$\tau_{cat} = \tau_2 + \tau_3 \left\{ \left(\frac{R_0}{R_0 + 1} \right) e^{-Fd_f/k_B T} + \left(\frac{1}{R_0 + 1} \right) e^{-F(d_f - d)/k_B T} \right\}^{-1}, \quad (3)$$

where τ_2 is $1/k_2$, and is independent of the load and τ_3 is $1/(k_{3f} + k_{3b})$ in the absence of a load. The dwell time decreases with the increment of the ATP concentration according to Michaelis–Menten kinetics (Nishiyama

Table 1
Rate constants of the branched kinetics of kinesin motor

Force (pN)	Energy difference of the barriers ^a ($k_B T$)	Rate constants ^b			
		k_1 ($\mu\text{M/s}$)	k_2 (s^{-1})	k_{3f} (s^{-1})	k_{3b} (s^{-1})
0	5.4	3.4	140	770	3.5
3.8	2.6	1.4	140	47	3.1
7.6	0	0.21	140	2.9	2.9

The parameters were calculated from the analysis of the dwell time for each force.

^a Energy difference = $5.4k_B T - 2.9F$ [nm].

^b $k_1 = k_{\text{cat}}/K_m$, $k_{\text{cat}} = k_2(k_{3f} + k_{3b})/(k_2 + k_{3f} + k_{3b})$, $k_2 = 1/\tau_2$, $k_{3f} = R_0 \exp(-Fd_f/k_B T)/(R_0 + 1)/\tau_3$, $k_{3b} = \exp(-Fd_b/k_B T)/(R_0 + 1)/\tau_3$, $K_m = 36 \mu\text{M}$ (0 pN) and $27 \mu\text{M}$ (3.8 and 7.6 pN), $\tau_2 = 7.0 \text{ ms}$, $\tau_3 = 1.3 \text{ ms}$, $R_0 = 221$, $d_f = 3.0 \text{ nm}$, $d_b = 0.1 \text{ nm}$.

et al., 2002). Thus, the dwell time, τ , is given by,

$$\tau = \tau_{\text{cat}} \left(1 + \frac{K_m}{[\text{ATP}]} \right), \quad (4)$$

where K_m is Michaelis constant ($=27 \mu\text{M}$), independent of the load. The average dwell time at ATP concentrations of 1 mM and 10 μM (Fig. 2d) could be well fitted by Eqs. (3) and (4) with values of $\tau_2 = 7.0 \text{ ms}$, $\tau_3 = 1.3 \text{ ms}$, and $d_f = 3.0 \text{ nm}$. The large characteristic distance of d_f is consistent with the first 4 nm sub-step (Fig. 2e). The parameter d_b was calculated to be 0.1 nm by $d_f - d$. The small value of d_b indicates that the backward barrier height is not significantly altered by the effect of a load. Table 1 shows the summary of the rate constants at 0, 3.8, and 7.6 pN (the stall force). The values are consistent with previous studies (Gilbert et al., 1995; Ma and Taylor, 1995; Higuchi et al., 1997; Kojima et al., 1997).

Our results can be summarized as follows. The stochastic behavior of the forward and backward movements can be explained simply by a Brownian motion of the mass particle biased in the forward direction and moving in a one-dimensional asymmetric potential. Our results show that the energy difference between the barrier heights in the forward and backward directions was $5.4k_B T$ ($T = 298 \text{ K}$) at zero load, which is about one fourth of the free energy of the ATP hydrolysis ($\sim 20k_B T$). The probability that movement occurred in the forward direction but not the backward, was load dependent and this is in agreement with similar assumption used in the Feynman's thermal ratchet model. In the next chapter, the stochastic behavior of the forward and backward movements using Feynman's model have been characterized.

3. Molecular motors and thermal ratchet mechanism

3.1. Feynman's thermal ratchet model

Oosawa introduced Feynman's thermal ratchet model to describe biased Brownian motion and analyzed the sliding motion of muscle fibers (Oosawa, 1989; Vale and Oosawa, 1990). Feynman's thermal ratchet model clearly states that the probabilities of movements in the forward and backward directions can be determined by the difference in temperature between the ratchet and the pawl (Feynman, 1963). In this chapter, the thermal ratchet model has been used to analyze the stepping motion of kinesin. The data of kinesin have also been compared with other processive molecular motors.

A molecular motor and protein track are supposed to be a ratchet and a series of pawls, respectively. The ratchet is shaped like a circular saw and it places between the pawls (Fig. 4c). One face of each pawl is oriented perpendicular to the circumference and the other inclines at a shallower angle. The ratchet and pawl are separated by different temperatures of T_f and T_b , respectively; in general, $T_f > T_b = T$ (experimental temperature). For the ratchet to move in the forward direction (towards the right in Fig. 4c) to the next site, it must acquire sufficient energy to withdraw the pawl, E , and work against the load, FL , where L is step size. The probability of obtaining this energy is proportional to $\exp(-(E + FL)/k_B T_f)$. It should be noted that in the model the load dependence of the probability is determined by the temperature, T_f , but not by the characteristic distance, d_f (Fig. 4b). The model is suitable for evaluating the external work

and work efficiency, because the effects of the work against the load are calculated by using the step size to a load-sensitive parameter.

The backward movement (towards the left in Fig. 4c) is inhibited by the block of the pawl, even if the ratchet has thermal energy, E . Movements in the backward direction, only occur if the pawl withdraws accidentally. The probability that this will occur is proportional to $\exp(-E/k_B T)$. The characterization of the thermal ratchet mechanism could be performed using the probabilities of the movements in the forward and backward directions.

The ratchet should be heated for making a certain temperature difference between the ratchet and pawl. For molecular motors, it may be performed by using the free-energy released from the hydrolysis of ATP molecules. But, the molecular motors spend a time for hydrolyzing a single ATP molecule and the ‘set-up’ time will differ from one motor to another; such reaction times are not essential for understanding the thermal ratchet mechanism. Thus, we have discussed only the rate constants of the stepping reactions in the forward and backward directions, k_{3f} and k_{3b} , but not k_1 and k_2 reactions (Fig. 4a).

3.2. Common temperature differences in molecular motors

According to the branched kinetics, the ratio of the movements in the forward and backward directions, k_{3f}/k_{3b} are described by,

$$\frac{k_{3f}}{k_{3b}} = R_1 \exp\left(-\frac{LF}{k_B T_f}\right), \quad (5)$$

$$R_1 = \exp\left[\frac{E}{k_B T} \left(\frac{T_f - T}{T_f}\right)\right]. \quad (6)$$

In the case of kinesin, the step size, $L = 8$ nm and the ratio of the forward to backward movement at ATP concentrations of 10 μ M and 1 mM have already been obtained (Fig. 3a). The data were fitted by Eqs. (5) and (6) with $T_f = 2.8T$, leading to $T_f = 834$ K at $T = 298$ K. The energy barrier in the potential was calculated to be $E = 8.4k_B T$, approximately half of the free energy of ATP hydrolysis.

The T_f for the other processive motors, 22S dynein of *Tetrahymena* cilia (Hirakawa et al., 2000), the inner-arm dynein C of *Chlamydomonas* flagella (Kojima

et al., 2002) and myosin V of chick brain (Mehta, 2001) has been calculated. The detailed analysis performed on kinesin was not completed for these motors, although the characteristic distance, d , and step size, L , have been measured by using SMD techniques. In equations of (1) and (5), d and L are used as coefficients of the external work against a load, respectively. There is essentially no difference between these equations. Thus, T_f is given by,

$$T_f = \frac{L}{d} T. \quad (7)$$

However, the following simple assumptions have been applied: $d_f = d$ and $d_b = 0$. Interestingly, the value of T_f for these motors was also about $3T$ (~ 900 K), which is almost the same as kinesin (Table 2). Vale and Oosawa (1990) obtained similar value by analyzing the experimental data of muscle fibers (myosin II). The ATP-driven molecular motors, kinesin, dynein, and myosin are likely to work using the same chemo-mechanical energy transduction mechanism. This is in agreement with the experimental results showing that the structure of the kinesin motor domain is similar not only to the other motors, myosin II and ncd, but also to G proteins (Kull et al., 1996; Sablin et al., 1996).

3.3. Functional diversity resulting from the potential barrier

Although these molecular motors have a similar temperature difference, the motors exhibit considerably different sliding velocities and maximum forces. Using the thermal ratchet mechanism with the same temperature difference for all motors, we demonstrate how these different characteristics could be produced. This demonstrates that nature carefully assigns the right ‘motor’ for the right ‘job’.

The thermal ratchet mechanism produces stepwise movements in both the forward and backward directions. The sliding velocity can be estimated by the relative probabilities of the ratchet undergoing forward versus backward movements.

$$V = V_0 \left[\exp\left(-\frac{E + FL}{k_B T_f}\right) - \exp\left(-\frac{E}{k_B T}\right) \right], \quad (8)$$

where V_0 is a constant, which determines the maximum speed caused by the sliding motion between the ratchet and pawl (Vale and Oosawa, 1990). When

Table 2
Mechanical characters of the ATP-driven motors

Motor molecules	Stall force (pN)	Step size (nm)	Characteristic distance (nm)	T_f/T	Barrier height ($k_B T$)
Kinesin	7.6	8.0	2.9	2.8	8.4
22S dynein	4.5	8.5	3.0	2.8	5.2
Inner-arm dynein C	1.2	8.3	3.3	2.5	1.6
Myosin V	3.0	36	10–15	3.0	8.8

Stall force, step size and characteristic distance were quoted from the following papers; 22S dynein (Hirakawa et al., 2000), inner-arm dynein C (Kojima et al., 2002) and myosin V (Mehta, 2001). In the experiment, single myosin V molecules generated force through three to five steps before stalling. The load before the final step in the forward direction was used to calculate the barrier height of myosin V. The value used for $F_{\max} = 2$ pN.

the load is increased sufficiently, the motor produces a maximum force ($F = F_{\max}$), and the velocity becomes zero. This occurs when

$$\frac{E + F_{\max} L}{k_B T_f} = \frac{E}{k_B T}. \quad (9)$$

Thus, the barrier energy can be obtained by rearranging this equation.

$$E = F_{\max} L \left(\frac{T}{T_f - T} \right). \quad (10)$$

The barrier energy is proportional to $F_{\max}$ and L , which have been measured using SMD techniques. By using the above equation and values of T_f and T , the barrier height of the molecular motors could be calculated (Table 2). Our results demonstrate that kinesin and myosin V are well designed for performing the external work, although these motors require large forces in order to change the direction of the movement. Kinesin and myosin V molecules in cells take ‘menial jobs’ such as vesicle transport, in which sudden changes in direction are not required.

In the absence of a load ($F = 0$), the velocity becomes,

$$V = V_0 \left[\exp \left(-\frac{E}{k_B T_f} \right) - \exp \left(-\frac{E}{k_B T} \right) \right]. \quad (11)$$

The probability that movements occur in the forward direction reaches to maximum. When $T_f > T$, the ratchet moves in the forward direction with a maximum velocity at a certain temperature difference. When $T_f = T$, the net velocity is zero, but it shows a one-dimensional diffusion. It has been reported that single one-headed kinesin shows thermal diffusion along a microtubule in the presence of ADP (Inoue et al., 2000; Okada and Hirokawa, 2000). In the presence of ATP, the movement was directional, indicating

that the free energy released from ATP hydrolysis is critical for creating temperature difference.

We calculated the sliding velocity at $T_f = 3T$, as a function of E by using Eq. (11) (Fig. 5). The peak is reached at $E = 1.65 k_B T$. Interestingly, this is the same as the value obtained for the potential depth of inner-arm dynein C (see arrow in Fig. 5). Distinct from the other motors that might work by single molecules, the inner-arm dynein C collaborates with seven distinct subspecies in *Chlamydomonas* flagella. The eight dynein molecules have the inherent oscillatory properties that are used in their generation of force and movement. In addition, Sakakibara et al. (1999) have reported that dynein C frequently slips backwards along a microtubule at high loads and this load-sensitive backward movement would facilitate to change the moving direction. If dynein molecules strongly bind to a microtubule, it would be hard to

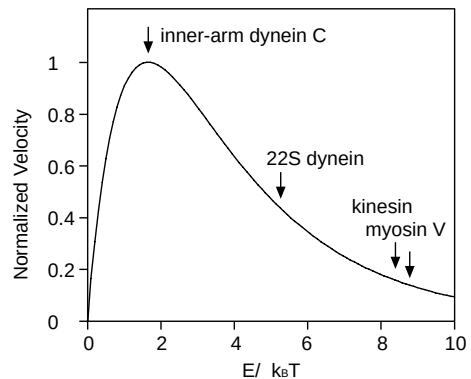


Fig. 5. The relation between the energy barrier and maximal velocity under zero load. The velocity was calculated from Eq. (11) with $T_f = 3T$ and has been normalized the peak value. Arrows indicate the barrier energy of each motor.

produce rhythmical oscillation of flagella in cooperation with the other motors. Therefore, the inner-arm dynein C may be a highly evolved molecular motor that produce the high speed sliding motion with similar underlying mechanism as other motors.

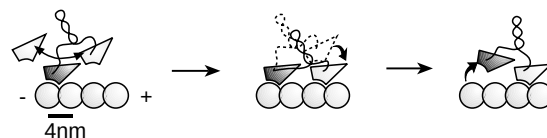
4. Stepping mechanism of kinesin

Here, we discuss the mechanism for the stepping movement taking into consideration the structural change coupled to the hydrolysis of ATP. As shown above, the hydrolysis reaction of single ATP molecules is coupled both to forward and backward movements. This indicates that the chemomechanical coupling is not deterministic. Thus, kinesin may be a loosely coupled motor, because the ATP hydrolysis reaction does not necessarily couple to the forward movement. How the forward and backward movements are created via ATP hydrolysis is outlined in Fig. 4a.

Kinesin has two motor domains called heads, each of them includes the ATP and microtubule-binding sites (Kull et al., 1996). In the absence of ATP, kinesin molecules attach to a microtubule via one head (rigor), leaving the other one free (Hirose et al., 1996; Uemura et al., 2002) (state [K]). ATP molecules bind to the head that is linked to the microtubule (state [K.T]). If the binding reaction of ATP molecules directly causes the stepwise movement, Michaelis constant, K_m , must be load dependent. However, the load dependency was not obtained (Nishiyama et al., 2002), and therefore [K] and [K.T] states are ‘immobile’ state.

The nucleotide hydrolysis process from ATP to ADP (state [K.D]) causes a structural change in the kinesin head (Rice et al., 1999; Kikkawa et al., 2001), such that the binding between kinesin and the microtubule is changed to the highly mobile state (Sosa et al., 2001). The energy released from ATP hydrolysis may be temporally stored in the kinesin–microtubule complex and this would be equivalent for the high-temperature state of motor molecules in the thermal ratchet model. Kinesin at this energy-charged state would have a chance to make an 8-nm step, however there would also be the risk of producing back steps and detachments. After kinesin molecules move to the next binding site, the ADP molecule would be released and the head would return to the immobile state at the new binding site (state [K]). This

Hand-over-hand



Hopping on one leg

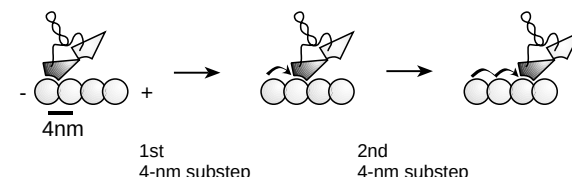


Fig. 6. Kinesin stepping models. The rate constant of the first 4 nm substep was dependent on the load, but the second rate constant was not, indicating that the 8-nm step was detected as the rapid two-step reactions irrespective of the loads.

rapid dissociation may play a role in inhibiting any additional movements, suggesting that each ATP hydrolysis produces a single stepwise movement. When this “stopper” function is removed, the multi-stepwise movements per ATP hydrolysis and the thermal diffusion can occur (Kitamura et al., 1999; Okada and Hirokawa, 2000).

The mechanism of the stepping motion is one of the central questions about the chemomechanical energy transduction of molecular motors, but as yet it has not been solved in spite of a great deal of effort. In the present section, we have presented several hypotheses. In the most widely accepted ‘hand-over-hand’ style model, the two heads of kinesin are assumed to work in coordinated manner (Vale and Milligan, 2000) (Fig. 6). A structural change in the motor domain initiates the partner head to attach to the next binding site, which is 8 nm from the initial binding site. The rear head would then detach from the initial binding site (the walking animation can be seen at the journal website <http://www.sciencemag.org/feature/data/1049155.shl>). Thus, the 8-nm step would be completed in two-step reactions, which is consistent with sequential 4-nm substeps (Fig. 2e). The role of each head would alternate at each ATPase cycle. The other model proposed is the walking model, i.e. ‘inchworm’ style movements, in which the roles do not change at each ATPase cycle (Hua

et al., 2002). These models propose that the coordinated action between two heads is critical for basal motility.

Recently, it has been reported that single one-headed kinesin molecules can move processively along a microtubule at zero load (Inoue et al., 2001; Okada and Hirokawa, 2000). This indicates that the two-headed structure is not required for motility, because a basal motility can be achieved by single head. To move for a considerable distance, such as the length of the motor domain (7 nm), the head attached to the microtubule might step along the 4-nm repeat of tubulin monomers in a way that resembles “hopping on one leg.” Then, the partner head might be to stabilize the kinesin–microtubule complex and assist in movements towards the plus end of the microtubule. The cooperation in both heads is critical for producing the large forces, because single one-headed kinesin was not able to produce the large forces (Okada et al., 2003). The direct observation of the motion of each head would shed light on the detailed mechanism on the force generation.

5. Conclusion and perspective

We developed a new assay to resolve nanometer displacements of single molecular motors with microsecond accuracy. The development allowed us to detect the individual stepwise movements. Single molecule analysis showed the biased Brownian motion mechanism on the stepwise movement. The driving force of the stepwise movement is essentially Brownian motion, but it is biased in the forward direction by using the free energy released from the hydrolysis of ATP. The mechanism used to bias the random movement can be well characterized by Feynman’s thermal ratchet model. It is likely that the free energy released from the hydrolysis of an ATP molecule is stored in several degrees of freedom in the molecules, as if these degrees were activated as a higher temperature (Oosawa, 2000). To prove this hypothesis, we have to detect the structural fluctuations by using the SMD techniques such as intermolecular force microscopy (Aoki et al., 2001), fluorescence resonance energy transfer imaging (Ishii et al., 1999) and diffracted X-ray tracking (Sasaki et al., 2001). Direct observation of the protein dynamics would of-

fer the key to an understanding on the mechanism of the biased Brownian motion.

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